

Acceptance Testing

UAT Execution & Report Submission

Date: 27 June 2026

Team ID: XXXXX

Project Name: Smart Lender

Maximum Marks: 4 Marks

Purpose of the Document

This document presents the **User Acceptance Testing (UAT)** conducted for the **Smart Lender – Loan Approval Prediction System**. The objective of the testing is to verify that the web application and Machine Learning model function correctly, satisfy the specified requirements, and provide accurate loan approval predictions. The report summarizes the executed test cases, test results, identified defects, and the overall acceptance status before project deployment.

Test Environment

Parameter	Details
Operating System	Windows 10/11
Development Environment	Visual Studio Code
Programming Language	Python 3.x
Framework	Flask
Frontend	HTML5, CSS3
Machine Learning Model	Random Forest Classifier (rdf.pkl)
Browser	Google Chrome / Microsoft Edge
Dataset	Loan Prediction Dataset

User Acceptance Test Cases

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
UAT-01	Open the Smart Lender application	Home page loads successfully	Home page displayed correctly	 Pass
UAT-02	Navigate to the prediction page	Prediction form opens	Form displayed successfully	 Pass

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
UAT-03	Enter valid applicant details	Data accepted without errors	Data submitted successfully	✓ Pass
UAT-04	Submit incomplete information	Validation message displayed	Validation worked correctly	✓ Pass
UAT-05	Predict loan approval	System generates prediction	Loan status displayed correctly	✓ Pass
UAT-06	Display prediction result	Approved/Rejected message shown	Result displayed instantly	✓ Pass
UAT-07	Test application responsiveness	Pages load properly on different screen sizes	Responsive layout verified	✓ Pass

Defect Report

Defect ID	Issue Identified	Severity	Resolution
D-01	Missing validation for empty fields	Medium	Added input validation in Flask backend
D-02	Incorrect input format handling	Low	Improved error handling and user messages
D-03	Minor UI alignment issue	Low	Updated CSS for consistent layout

Performance Testing Summary

Parameter	Result
Application Startup	Successful
Form Submission Time	Less than 2 seconds
Prediction Response Time	Less than 1 second
Accuracy of Predictions	High (based on trained Random Forest model)
Application Stability	Stable during testing

User Feedback

- The application is simple and easy to use.
- Loan prediction results are generated quickly.
- The user interface is clean and intuitive.

- Input validation helps prevent incorrect data entry.
- Overall experience is satisfactory and suitable for demonstration purposes.

Final UAT Summary

- **Total Test Cases Executed:** 7
- **Passed:** 7
- **Failed:** 0
- **Defects Identified:** 3 (Minor)
- **Defects Resolved:** 3
- **Final Status:** Accepted for Deployment

Conclusion

The **Smart Lender** application successfully passed User Acceptance Testing. All major functional requirements were verified, the Random Forest Machine Learning model produced accurate loan approval predictions, and the web application performed reliably during testing. Minor issues identified during testing were resolved, making the application ready for final demonstration and deployment.

This version is more complete, professional, and appropriate for your internship documentation; while matching the features you've actually implemented in your **Smart Lender** project.